

PAP521S - PROGRAMMING IN PRACTICE

PROJECT A: MUNICIPAL FINANCIAL MANAGEMENT SYSTEM

INDIVIDUAL CONTRIBUTION RECORD

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Assigned Responsibility: Role 2 - Budget Management Module

1. Overview

I was responsible for the Budget Management module of the Municipal Financial Management System (MFMS). It tracks how much money each department was given compared with how much it has spent, and flags any department that has overspent.

2. Files Created

- budget.c - contains the code for the module.
- budget.h - declares the functions used by other files.

3. Tasks Completed

- Enter departmental budgets and expenditure.
- Calculate the remaining budget (budget minus expenditure).
- Determine the status of each department: WITHIN BUDGET or EXCEEDED.
- Display the budget information for all departments.
- List the departments that overspent.
- Input validation (no negative budget or expenditure).

4. Integration with Other Modules

- The Budget Management option in main.c calls the module's menu function.
- Budget data can be used by the Reports module, for example to list the departments that went over budget.

5. Testing

- Entered several departments and checked that the budget information was displayed correctly.
- Checked that the remaining budget is calculated correctly.
- Checked that a department over budget shows EXCEEDED and one within budget shows WITHIN BUDGET.
- Checked that the list of overspent departments contains only the departments that exceeded their budgets.
- Entered a negative budget and a negative expenditure to confirm they are rejected.

6. Conclusion

My contribution gives the MFMS a reliable way to track departmental budgets and spending and to identify departments that have overspent.